

Section 4.4

Exercise (4.4.1). If $\sigma \in \text{Aut}(G)$ and φ_g is conjugation by g prove $\sigma\varphi_g\sigma^{-1} = \varphi_{\sigma(g)}$. Deduce that $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$.

Proof: Suppose $\sigma \in \text{Aut}(G)$ and φ_g is conjugation by $g \in G$. Consider the map $\sigma\varphi_g\sigma^{-1}(a)$ for some $a \in G$ with the property that $\sigma^{-1}(a) = b$. Then by definition we get that

$$\sigma\varphi_g\sigma^{-1}(a) = \sigma\varphi_g(b) = \sigma(gbg^{-1}).$$

Finally by properties of isomorphisms we get that $\sigma(g)\sigma(b)\sigma(g^{-1}) = \sigma(g)a\sigma(g)^{-1}$.

Thus $\sigma\varphi_g\sigma^{-1} = \varphi_{\sigma(g)}$. It follows directly that $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$ since we have shown that for all $\sigma \in \text{Aut}(G)$ and $\varphi_g \in \text{Inn}(G)$, then $\sigma\varphi_g\sigma^{-1} = \varphi_{\sigma(g)}$. We know $\sigma(g) \in G$ and therefore $\varphi_{\sigma(g)} \in \text{Inn}(G)$ thus by definition $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$. $\square$

Exercise (4.4.6). Prove that characteristic subgroups are normal. Give an example of a normal subgroup that is not characteristic.

Proof: Suppose $H \text{ Char } G$. Therefore by definition for all $\sigma \in \text{Aut}(G)$ we know that $\sigma(H) = H$. In the previous exercise we deduced that $\text{Inn}(G) \trianglelefteq \text{Aut}(G)$. Therefore for all $\phi \in \text{Inn}(G)$, $\phi(H) = gHg^{-1} = H$ for all $g \in G$. $\square$

Exercise (4.4.7). If H is the unique subgroup of a given order in a group G prove H is characteristic in G .

Proof: Suppose $H \leq G$ with $|H| = p$ such that p is unique in the set of all subgroup orders, and H is not characteristic in G . By definition there exists some $\sigma \in \text{Aut}(G)$ such that $\sigma(H) \neq H$. Note that since σ is an isomorphism, so $|\sigma(H)| = |H|$ is a contradiction. $\square$

Exercise (4.4.8). Let G be a group with subgroups H and K with $H \leq K$

a. Prove that if H is characteristic in K and K is normal in G then H is normal in G .

Proof: Suppose that H is characteristic in K and K is normal in G . By definition we know that for all $g \in G$, $gKg^{-1} = K$. Said another way we know that for

all $\phi \in \text{Inn}(G)$ we get $\phi(K) = K$. Note that since $\phi(K) = K$ we know that $\phi|_K \in \text{Aut}(K)$. Since H is characteristic in K , by definition for all $\sigma \in \text{Aut}(K)$ $\sigma(H) = H$. Thus $\phi|_K(H) = H$ and since $H \leq K$, $\phi(H) = H$ thus $H \trianglelefteq G$.

□

- b. Prove that if H is characteristic in K and K is characteristic in G then H is characteristic in G . Use this to prove that the Klein 4-group V_4 is characteristic in S_4 .

Proof: Suppose $H \text{ Char } K$, and $K \text{ Char } G$. By definition for all $\sigma_g \in \text{Aut}(G)$ we know that $\sigma_g(K) = K$. Note that $\sigma_g|_K \in \text{Aut}(K)$. We also have by definition that for all $\sigma_k \in \text{Aut}(K)$, $\sigma_k(H) = H$. Therefore $\sigma_g|_K(H) = H$ and since $H \leq K$ we get that $\sigma_g(H) = H$ thus $H \text{ Char } G$.

We will show that V_4 is characteristic in S_4 by showing that $V_4 \text{ Char } A_4$ and $A_4 \text{ Char } S_4$. Note that since automorphisms preserve element order, for all $\sigma \in \text{Aut}(S_4)$, $\sigma(A_4)$ must map $\sigma(\langle(123), (124), (134), (234)\rangle) = \langle(123), (124), (134), (234)\rangle$ and the identity permutations must map together. This leaves only 3 elements of order 2 in A_4 left to map therefore it must be the case that $\sigma(\langle(12)(34), (13)(24)\rangle) = \langle(12)(34), (13)(24)\rangle$ (it can't be either the transpositions or the 4-cycles as we can only map 3 elements and those map in pairs). Thus for all $\sigma \in \text{Aut}(S_4)$, $\sigma(A_4) = A_4$. Note that $V_4 = \langle(12)(34), (13)(24)\rangle$, again since automorphisms preserve element order, for all $\sigma \in \text{Aut}(A_4)$, $\sigma(\langle(12)(34), (13)(24)\rangle) = \langle(12)(34), (13)(24)\rangle$ and identity permutations must map. Thus for all $\sigma \in \text{Aut}(A_n)$, $\sigma(V_4) = V_4$. Thus by the previous conclusion V_4 is characteristic in S_4 .

□

- c. Give an example to show that if H is normal in K and K is characteristic in G then H need not be normal in G .

Solution: We have just shown that $V_4 \text{ Char } A_4$. Consider $H = \langle(12)(34)\rangle$ and note that $H \leq V_4$. Note that $H \trianglelefteq V_4$, since every $(ab)(cd)$ is its own inverse we see that $(ab)(cd)(12)(34)(ab)(cd) = (12)(34)$. However H is not normal in S_4 , consider $(24)(12)(34)(24) = (14)(23)$.

□

Exercise (4.4.9). If r, s are the usual generators for the dihedral group D_{2n} , use the preceding two exercise to deduce that every subgroup of $\langle r \rangle$ is normal in D_{2n} .

Proof: Suppose r, s are the usual generators for the dihedral group D_{2n} and consider the subgroup $\langle r \rangle$. Recall Theorem 7 of Section 2.3 which stated if a cyclic group $\langle x \rangle$ has finite order n the subgroups of $\langle x \rangle$ correspond bijectively with the positive divisors of n . Therefore every subgroup of $\langle r \rangle$ has unique order and is therefore characteristic in $\langle r \rangle$. Also note that $|D_{2n} : \langle r \rangle| = 2$, so there are two cosets $\langle r \rangle, g\langle r \rangle$ for $g \in D_{2n}$. Therefore $g\langle r \rangle = \langle r \rangle g$, and thus $\langle r \rangle \trianglelefteq D_{2n}$. By the previous exercise we get that every subgroup of $\langle r \rangle$ is normal in D_{2n} . $\square$

Exercise (4.4.12). Let G be a group of order 3825. Prove that if H is a normal subgroup of order 17 in G then $H \leq Z(G)$.

Proof: Suppose G is a group of order 3825, and $H \trianglelefteq G$ with $|H| = 17$. Note that since H has prime order it must also be a cyclic subgroup, and recall that since H is normal by definition we know that $N_G(H) = G$. Consider Corollary 15, which tells us that the quotient group $N_G(H)/C_G(H) \cong S_H$ where $S_H \leq \text{Aut}(H)$. Applying Proposition 16 we find out that since H is cyclic, $\text{Aut}(H) \cong (\mathbb{Z}/(17)\mathbb{Z})^*$ with $|\text{Aut}(H)| = \varphi(17) = 16$. Applying Lagrange's Theorem we get that $|S_H| \mid |\text{Aut}(H)| = 16$ and therefore $|N_G(H)|/|C_G(H)| \mid 16$. By substitution we get $3825/|C_G(H)| \mid 16$ and since no divisor of 16 other than 1 divides 3825, $|N_G(H)|/|C_G(H)| = 1$ and therefore $|C_G(H)| = 3825$ thus $C_G(H) = G$ which implies that $H \leq Z(G)$. $\square$

Section 4.5

Exercise (4.5.1). Prove that if $P \in \text{Syl}_p(G)$ and H is a subgroup of G containing P then $P \in \text{Syl}_p(H)$. Give an example to show that, in general, a Sylow p -subgroup of a subgroup of G need not be a Sylow p -subgroup of G .

Proof: Suppose $P \in \text{Syl}_p(G)$ and H is a subgroup of G containing P . By definition $|G| = p^\alpha m$ for prime p , not dividing m . Since $P \in \text{Syl}_p(G)$ we know that $|P| = p^\alpha$.

Since $H \leq G$ and $P \leq H$ by Lagrange's Theorem we know that $|H| \mid |G|$ and $|P| \mid |H|$. Therefore it must follow that $|H| = p^\alpha j$ where $j \leq m$. Thus by definition $P \in \text{Syl}_p(H)$.

For an example consider S_4 , A_4 , and V_4 . Note that V_4 is a Sylow 2-subgroup of A_4 . Note that since $|S_4| = 2^3 \cdot 3$ any Sylow 2-subgroup of S_4 will be order $2^3 = 8$, so $V_4 \notin \text{Syl}_p(A)$. $\square$

Exercise (4.5.14). Prove that a group of order 312 has a normal Sylow p -subgroup for some prime p dividing its order.

Proof: Suppose that G has order 312. Consider the prime factorization of $312 = 2^3(3)(13)$.

Therefore in G there exists a Sylow 2-subgroup order 2^3 , a Sylow 3-subgroup order 3, and a Sylow 13-subgroup order 13. Recall Sylow's Theorem, which states that n_{13} , the number of Sylow 13-subgroups divides $2^3(3) = 26$ and has the form $n_{13} = 1 + k(13)$. Any $k > 1$ will result in $n_{13} > 26$. With $k = 1$ we get $n_{13} = 14$ which does not divide 26. Therefore $k = 0$ so $n_{13} = 1$. Again by Sylow's Theorem we get that for our Sylow 13-subgroup P , $|G : N_G(P)| = 1$ which means that $N_G(P) = G$ thus $P \trianglelefteq G$. $\square$

Exercise (4.5.25). Prove that if G is a group of order 385 then $Z(G)$ contains a Sylow 7-subgroup of G and a Sylow 11-subgroup is normal in G .

Proof: Suppose G has order 385. Consider the prime factorization, $385 = (5)(7)(11)$. Therefore in G there exists a Sylow 5-subgroup order of order 5, a Sylow 7-subgroup P_7 , of order 7 and a Sylow 11-subgroup P_{11} , of order 11. From Sylow's Theorem we know that n_7 divides $(11)(5) = 55$, and it has form $n_7 = 1 + k(7)$. Note that $k > 7$ will not divide 55. Consider the following table which demonstrates that for $0 < k \leq 7$, n_7 does not divide 55.

$k(7)$	0	7	14	21	28	35	42	49
n_7	1	8	15	22	29	36	43	50
remainder	0	1	5	11	3	17	32	45

Thus $n_7 = 1$ and similarly to the previous problem $P_7 \trianglelefteq G$. Therefore $N_G(P_7) = G$ and recall that the quotient group $N_G(P_7)/C_G(P_7) \cong S_{P_7}$ for some $S_{P_7} \in \text{Aut}(P_7)$. Since P_7 is cyclic from Proposition 16 of Section 4.4 we know that $|\text{Aut}(P_7)| = \varphi(17) = 16$. Applying Lagrange's Theorem we know that $S_{P_7} \mid 16$ and therefore $|N_G(P_7)|/|C_G(P_7)| \mid 16$. Since P_7 is normal we get $385/|C_G(P_7)| \mid 16$, and since and since no divisor of 16 other than 1 divides 385 we get that $|N_G(P_7)|/|C_G(P_7)| = 1$ so therefore $|C_G(P_7)| = 385$ which implies that $C_G(P_7) = G$. Thus $P_7 \in Z(G)$.

From Sylow's theorem we know that n_{11} divides $(5)(7) = 35$, and it has the form $n_{11} = 1 + k(11)$. Clearly $12 \nmid 35$, $23 \nmid 35$ and $34 \nmid 35$, so $n_{11} = 1$ and therefore $P_{11} \trianglelefteq G$. □

Exercise (4.5.26). Let G be a group of order 105. Prove that if a Sylow 3-subgroup of G is normal then G is abelian.

Proof: Suppose $|G| = 105$ with a Sylow 3-subgroup P_3 that is normal in G . Since P_3 is normal we know that $N_G(P_3) = G$. Consider the factorization $105 = (3)(5)(7)$, so $|P_3| = 3$ and is therefore cyclic. Recall that the quotient group $N_G(P_3)/C_G(P_3) \cong S_{P_3}$ for some $S_{P_3} \in \text{Aut}(P_3)$. Since P_3 is cyclic from Proposition 16 of Section 4.4 we know that $|\text{Aut}(P_3)| = \varphi(3) = 2$. Applying Lagrange's Theorem we know that $S_{P_3} \mid 2$ and therefore $|N_G(P_3)|/|C_G(P_3)| \mid 2$. Since $2 \nmid 105$ we get that $|N_G(P_3)|/|C_G(P_3)| = 1$ so therefore $|C_G(P_3)| = 105$ which implies that $C_G(P_3) = G$. Thus $P_3 \in Z(G)$.

Now consider the quotient group $G/Z(G)$. By Lagrange's Theorem, since $|G/Z(G)| = |G|/|Z(G)|$ and $|Z(G)| \geq 3$ since $P_3 \leq Z(G)$ we know that $|G/Z(G)|$ is either 35, 7, 5 or 1. In case that $|G/Z(G)|$ is 7, 5 or 1 we get that $G/Z(G)$ is cyclic and therefore G is abelian.

Suppose $|G/Z(G)| = 35$. Therefore $G/Z(G)$ has a Sylow 5-subgroup of order 7. From Sylow's Theorem we know that n_5 divides 7 and it has from $n_5 = 1 + k(5)$. Therefore it follows that $k = 0$ and $n_5 = 1$. Thus P_5 is normal in $G/Z(G)$. Therefore

$G/Z(G)$ is a pq subgroup with $P_5 \trianglelefteq G/Z(G)$, by the Groups of order pq example we know that $G/Z(G)$ must be cyclic and G is abelian. $\square$

Exercise (4.5.36). Prove that if N is a normal subgroup of G then $n_p(G/N) \leq n_p(G)$.

Proof: Suppose that N is a normal subgroup with order $p^\beta n$ and G has order $p^\alpha m$. Consider the quotient group G/N , By Lagrange's Theorem $|G/N| = p^{\alpha-\beta} m/n$. Let P be a Sylow p -subgroup. By Sylow's Theorem we know $n_p(G/N)$ divides m/n , and $n_p(G)$ divides m . $\square$

Section 4.6

Exercise (4.6.3). Prove that A_n is the only proper subgroup of index $< n$ in S_n for all $n \leq 5$.

Proof: $\square$